

**Department of Electrical Engineering
The Superior University, Lahore**

Computer Programming

Experiment No.1

Introduction to Algorithms

Prepared for:

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By:

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Section: Biomedical Engineering

Semester: 2nd

Total Marks: 10

Obtained Marks:

Signature:

Date:

Experiment No.1
Introduction to Algorithms
(Rubrics)

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A. PSYCHOMOTORp

Serial No.	Criteria	Allocated Marks	Unacceptable 0%	Poor 25%	Fair 50%	Good 75%	Excellent 100%	Total Obtained
1	Follow Procedures	2	0	0.5	1	1.5	2	
2	Designing an Algorithm	4	0	0.5	1	2.5	4	
Sub-Total		6	Sub Total Marks Obtained in Psychomotor (P)					

B. AFFECTIVE

Serial No.	Criteria	Allocated Marks	Unacceptable 0%	Poor 25%	Fair 50%	Good 75%	Excellent 100%	Total Obtained
1	Respond to Questions	2	0	0.5	1	1.5	2	
2	Lab Report	2	0	0.5	1	1.5	2	
Sub-Total		4	Sub Total Marks Obtained in Affective (A)					

Instructor Name: Miss Eisha NawazTotal Marks (P+A): 10

Instructor Signature: _____

Obtained Marks (P+A): _____

CONTENTS OF LAB 1

1	Objective:	4
2	Introduction to Algorithm	4
2.1	Example 1	4
2.2	Example 2	5
2.3	Example 3	5
2.4	Example 4	5
2.5	Example 4	6
2.6	Example 6	6
2.7	example 7	7
2.8	example 8	7
3	Lab Tasks	9
3.1	Task 1	9
3.2	Task 2	9
3.3	Task 3	9
3.4	Task 4	9
3.5	Task 5	9
4	Home Assignment	10
4.1	Task 1	10
4.2	Task 2	10

Introduction to Algorithm

1 OBJECTIVE:

To briefly introduce of algorithm for defining and describing a big problem in small parts, so that it is very easy to execute. Its finite sequence of step to solve a problem.

2 INTRODUCTION OF ALGORITHM

An algorithm is a process or set of rules which must be followed to complete task. This is basically the step-by-step procedure to complete any task.

- An algorithm should be basic and easy to perform.
- Each step started with a specific indentation like, “Step-1”,
- There must be “Start” as the first step and “End” as the last step of the algorithm.

There are some basics steps to make an algorithm:

1. **Start** – Start the algorithm
2. **Input** – Take the input for values in which the algorithm will execute.
3. **Conditions** – Perform some conditions on the inputs to get the desired output.
4. **Output** – Printing the outputs.
5. **End** – End the execution.

Let's take some examples of algorithms for computer science problems.

2.1 USE OF THE ALGORITHMS:

Algorithms play a crucial role in various fields and have many applications. Some of the key areas where algorithms are used include:

Computer Science: Algorithms form the basis of computer programming and are used to solve problems ranging from simple sorting and searching to complex tasks such as artificial intelligence and machine learning.

Mathematics: Algorithms are used to solve mathematical problems, such as finding the optimal solution to a system of linear equations or finding the shortest path in a graph.

Operations Research: Algorithms are used to optimize and make decisions in fields such as transportation, logistics, and resource allocation.

Artificial Intelligence: Algorithms are the foundation of artificial intelligence and machine learning, and are used to develop intelligent systems that can perform tasks such as image recognition, natural language processing, and decision-making.

Data Science: Algorithms are used to analyze, process, and extract insights from large amounts of data in fields such as marketing, finance, and healthcare.

2.2 EXAMPLE 1

The given example shows the algorithm of finding the area of a rectangle:

Table 1 Example 1

Step 1:	Start
Step 2:	Take the Height and Width of the rectangle as input.
Step 3:	Declare a variable as “area”
Step 4:	Multiply Height and Width
Step 5:	Store the multiplication to “Area”, (its look like area = Height x Width)
Step 6:	Print “area”;
Step 7:	End

2.3 EXAMPLE 2

The given example shows the algorithm of finding the area of a circle:

Table 2 Example 2

Step 1:	Start
Step 2:	Declare variable R, P, AREA
Step 3:	Read value R
Step 4:	Set P=3.14
Step 5:	AREA=P*R*R_____
Step 6:	Print Area

Step 6:	End
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2.4 EXAMPLE 3

The given example shows the algorithm of finding the greatest between 3 numbers:

Table 3 Example 3

Step 1:	Start
Step 2:	Take 3 numbers as input, say A, B, and C.
Step 3:	Check if(A>B and A>C)
Step 4:	Then A is greater
Step 5:	Print A
Step 6:	Else
Step 7:	Check if(B>A and B>C)
Step 8:	Then B is greater
Step 9:	Print B
Step 10:	Else C is greater
Step 11:	Print C
Step 12:	End

2.5 EXAMPLE 4

The given example shows the algorithm of adding two numbers entered by the user:

Table 4 Example 4

Step 1:	Start
Step 2:	Declare variables num1, num2 and sum.
Step 3:	Read values num1 and num2
Step 4:	Add num1 and num2 and assign the result to sum. $Sum = num1 + num2$
Step 5:	Print sum
Step 6:	End

2.6 EXAMPLE 4

The given example shows the algorithm of finding whether a given number is odd or even:

- Odd numbers are the numbers that are not divisible by 2, it starts from 1. In other words, these numbers leave 1 as a remainder.

- Even numbers are the numbers that are divisible by 2, it starts from 0. In other words, these numbers leave 0 as a remainder.

Table 5 Example 5

Step 1:	Start
Step 2:	Declare variable n, remainder.
Step 3:	Read number n
Step 4:	Find the remainder of the number divided by 2 using Remainder = n % 2
Step 5:	Check: If remainder == 0, then display "Even".
Step 6:	Else
Step 7:	Display "ODD"
Step 8:	End

2.7 EXAMPLE 6

The given example shows the algorithm of swapping the value of 2 integers:

First way is to store one of the variable value, say A's value in another memory location, a temporary third variable T. Then copy B's value into A and then T's value into A. In this approach we need an additional storage to store the value of a number.

Table 6 Example 6

Step 1:	Start
Step 2:	Declare variable A, B, N.
Step 3:	Input A, B.
Step 4:	N = A; A = B; B = N;
Step 5:	Print A
Step 6:	Print B
Step 7:	End

2.8 EXAMPLE 7

The given example shows the algorithm to find out the profit or loss of product:

Table 7 Example 7

Step 1:	Start
Step 2:	Input product price as pp

Step 3:	<i>Input selling price as sp</i>
Step 4:	<i>If pp < sp Print profit as pp - sp</i>
Step 5:	<i>Else If sp < pp Print loss As sp-pp</i>
Step 6:	<i>If pp = sp Print no profit no loss</i>
Step 7:	End

2.9 EXAMPLE 8

The given example shows the algorithm to find the grade of student according to his/her marks:

Table 8 Example 8

Step 1:	Start
Step 2:	<i>Declare variable n, obtained marks, total marks</i>
Step 3:	<i>Input obtained marks, total marks</i>
Step 4:	<i>$n = (\text{obtained marks} / \text{total marks}) * 100$</i>
Step 5:	<i>If n > 80 Print grade A+</i>
Step 6:	<i>If n > 70 and n < 80 Print grade A</i>
Step 7:	<i>Else If n > 60 and n < 70 Print grade B</i>
Step 8:	<i>Else If n > 50 and n < 60 Print grade C</i>
Step 9:	<i>Else If n > 40 and n < 50 Print grade D</i>
Step 10:	<i>Else If n > 33 and n < 40 Print grade E</i>
Step 11:	<i>Else Print grade F</i>
Step 12:	End

3 LAB TASKS

Solve the following tasks.

3.1 TASK 1

Write an algorithm where the science, math and English marks of a student are input. Calculate the average percentage. If the percentage is above 40 then display passed else display failed.

```

Step 1 : Start
Step-2 : Declare variables science, maths, english,
Step 3 : Input the marks of science, maths, english
Step 4 : Calculate total = science + maths + english
Step 5 : Calculate percentage average = (total / 3) * 100 %
Step 6 : If percentage > 40
           display passed
Step 7 : If percentage < 40
           display failed
Step 8 : End
  
```

3.2 TASK 2

Write an algorithm of finding the Area of the Right-angle Triangle.

Hint:

$$\text{Area} = \frac{1}{2}(\text{base} \times \text{height})$$

Task - 2

```

Step 1: Start
Step 2: Take base and height as input
Step 3: Declare variable area
Step 4: area =  $\frac{1}{2} \times \text{base} \times \text{height}$ 
Step 5: Print area
Step 6: End
  
```

3.3 TASK 3

Write an algorithm where the temperature in centigrade is entered. If the input temperature is more than 30 degrees, display “it is warm today”, else display “It is cool today”.

Task - 3

Step 1 : Start

Step 2 : Input temperature in °C.

Step 3 : If temperature > 30

Display 'It is warm today'

Step 4 : If temperature < 30

Display 'It is cool today'

Step 5 : End

3.4 TASK 4

Write an algorithm where the user entered the total number of umbrellas, he/she wants to buy. Price of one umbrella is Rs 400. Calculate the total price of umbrella user purchase and display the result.

Task - 4

Step 1 : Start

Step 2 : Input number of umbrellas n

Step 3 : Set price of an umbrella = 400 Rs

Step 4 : Calculate total price = $n \times 400$ Rs

Step 5 : Display total

Step 6 : End

3.5 TASK 5

Write an algorithm of finding the sum and the average of three numbers and display the sum and average separately.

Task - 5

Step 1 : Start

Step 2 : Input three numbers, say a, b, c

Step 3 : Calculate sum

$$\text{sum} = a + b + c$$

Step 4 : Calculate average

$$\text{average} = \text{sum} / 3$$

Step 5 : Display sum

Step 6 : Display average

Step 7 : End

4 HOME ASSIGNMENT**4.1 TASK 1**

Write an algorithm to find the circumference of a circle with a given radius.

Hint:

$$C = 2\pi r$$

Task - 1

Step 1 : Start

Step 2 : Input radius r

Step 3 : Calculate circumference

~~Step 4~~ : circumference = $2 \times \pi \times r$ ($2 \times 3.14 \times r$)

Step 4 : Display circumference

Step 5 : End

4.2 TASK 2

Write an algorithm to check the temperature, if the temperature in Fahrenheit convert it in Celsius, and if the temperature in Celsius convert it in Fahrenheit. Display the final temperature.

Task - 2

Step 1 : Start

Step 2 : Input temperature value T

Step 3 : Input type (°C/°F)

Step 4 : If unit = °C, convert to fahrenheit. °F.
$$^{\circ}\text{F} = \left(\frac{9}{5} \times T\right) + 32$$

§ - Display °F

Step 5 : Else if unit = °F, convert to celsius °C.

$$^{\circ}\text{C} = \left(\frac{5}{9} \times (T - 32)\right)$$

- Display °C

Step 6 : End